# Creatinine

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## Identity Sheet

| Property | Value |
|---|---|
| **Name** | Creatinine |
| **Other names** | 2-Amino-1-methyl-2-imidazolidin-4-one, Creatinin |
| **Chemical formula** | $C_4H_7N_3O$ |
| **Molecular weight** | 113.12 g/mol |
| **CAS number** | 60-27-5 |
| **PubChem CID** | 588 |
| **SMILES** | CN1CC(=O)NC1=N |
| **Appearance** | White crystalline powder |
| **Melting point** | 300 C (decomposes) |
| **Solubility in water** | ~90 g/L at 20 C |

**Structural formula:**

```
CH3
|
N—CH2
| |
C=NH C=O
|
NH
```

Creatinine is a cyclic derivative of creatine, formed by spontaneous non-enzymatic dehydration of creatine phosphate in muscle. It is the most widely used endogenous marker of glomerular filtration rate (GFR). Its production rate is relatively constant for a given individual, proportional to muscle mass.

### Molecules Not to Be Confused With

| Molecule | Formula | MW (g/mol) | Key Difference |
|---|---|---|---|
| **Creatine** | $C_4H_9N_3O_2$ | 131.13 | Precursor; amino acid derivative; not a waste product |
| **Creatine phosphate** | $C_4H_{10}N_3O_5P$ | 211.11 | Energy storage molecule in muscle; dephosphorylation yields creatinine |
| **Urea** | $CH_4N_2O$ | 60.06 | Different nitrogen waste product; from amino acid catabolism, not muscle |

| Molecule | Formula | MW (g/mol) | Key Difference |
|---|---|---|---|
| **Cystatin C** | Protein | 13,343 | Alternative GFR marker; not affected by muscle mass |

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## Medical Information

### Origin

***Endogenous: Biosynthesis – Creatine-Phosphocreatine Pathway*** Creatinine is produced by **irreversible, non-enzymatic cyclisation** of creatine phosphate (and to a lesser extent, free creatine) in skeletal muscle. The process:

1. **Creatine synthesis:** Arginine + Glycine (in kidney) via AGAT -> guanidinoacetate, then methylated by GAMT (in liver) using SAM -> **creatine**.
2. **Phosphorylation:** Creatine + ATP (via creatine kinase) -> **creatine phosphate** (PCr) in muscle.
3. **Creatinine formation:** PCr spontaneously dehydrates at ~1.7%/day -> **creatinine** (irreversible).

Daily creatinine production: ~15-25 mg/kg/day (~1.0-1.8 g/day), proportional to muscle mass. Production is remarkably constant day-to-day for an individual.

***Exogenous*** Dietary creatinine from cooked meat (~1.5-2.5 g creatine per kg of meat; cooking converts some creatine to creatinine). Creatine supplements increase creatinine production.

### Primary & Secondary Biological Roles

**Primary role:** - **Waste product / GFR marker:** Creatinine has no physiological function. Its clinical value lies in being a nearly ideal endogenous GFR marker: freely filtered, not reabsorbed, minimal secretion (10-15% tubular secretion).

**Secondary roles:** - No known biological function. - Used as a normalisation factor for spot urine measurements of other analytes (analyte/creatinine ratio corrects for urine concentration).

### Catabolism and Elimination Pathway

- Creatinine is released from muscle into plasma at a constant rate.
- **Freely filtered** at the glomerulus (100%).
- ~10-15% **secreted** by proximal tubular organic cation transporters (OCT2, MATE1/2).
- **Not reabsorbed** (or minimally so).
- Urinary creatinine excretion = creatinine production rate at steady state.
- Creatinine clearance (CrCl) approximates GFR (slight overestimate due to secretion).

### Expression in Humans

***Normal Levels***

| Compartment | Reference Range |
|---|---|
| **Serum creatinine** | Male: 0.7-1.3 mg/dL (62-115 umol/L); Female: 0.6-1.1 mg/dL (53-97 umol/L) |
| **Urinary creatinine (24-h)** | Male: 14-26 mg/kg/day (1.0-1.8 g/day); Female: 11-20 mg/kg/day (0.8-1.4 g/day) |
| **Urinary creatinine concentration** | 3-30 mmol/L (highly variable with hydration) |

| Compartment | Reference Range |
|---|---|
| **Creatinine clearance** | 90-140 mL/min (approximates GFR) |
| **eGFR (CKD-EPI)** | >90 mL/min/1.73 m2 |

***Factors Influencing Levels*** **Increased serum creatinine (decreased GFR):** - Acute kidney injury (AKI) - Chronic kidney disease (CKD) - Rhabdomyolysis (massive creatine release) - High meat diet (transient) - Creatine supplements - Drugs inhibiting secretion (trimethoprim, cimetidine) – raise serum Cr without changing GFR

**Decreased serum creatinine:** - Low muscle mass (cachexia, elderly, amputees) - Liver disease (reduced creatine synthesis) - Pregnancy (increased GFR + hemodilution)

***Associated Pathologies***

| Condition | Creatinine Pattern | Key Symptoms |
|---|---|---|
| **AKI** | Rapidly rising serum Cr (>0.3 mg/dL in 48 h) | Oliguria, fluid overload |
| **CKD** | Chronically elevated serum Cr | Fatigue, anaemia, bone disease |
| **Rhabdomyolysis** | Very high serum Cr + CK | Muscle pain, dark urine, AKI risk |
| **Muscle wasting** | Low serum Cr, low urinary Cr | Sarcopenia, malnutrition |

### Presence in Urine

**Should it be normally present? Yes** – creatinine is a major normal urinary solute. It is the basis for urine concentration normalisation.

**Normal urinary levels:** 1.0-1.8 g/day (8.8-15.9 mmol/day) for men; 0.8-1.4 g/day for women. Concentration: 3-30 mmol/L.

**Form in urine: Native dissolved molecule** (neutral at urine pH, pKa ~4.8 for the imino group).

**Pathological significance:**

| Urinary Creatinine | Possible Causes | Prevalence |
|---|---|---|
| **Low 24-h excretion** | Incomplete collection, low muscle mass, advanced CKD | Common artefact |
| **High 24-h excretion** | High muscle mass, meat-rich diet | Normal variant |

**Solubility:** ~90 g/L at 20 C. Creatinine does not crystallise in urine normally.

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## Detection in Urine

### Available Clinical Assays

1. **Jaffe reaction (alkaline picrate, most common):**
   - **Principle:** Creatinine reacts with picric acid in alkaline solution -> orange-red Janovsky complex.
   - **Detection:** Absorbance at **520 nm** (kinetic measurement).
   - **LOD:** ~0.1 mg/dL (~9 umol/L).
   - **Advantages:** Simple, inexpensive, widely available on all platforms.

- **Disadvantages:** Non-specific; interfered by glucose, acetoacetate, proteins, cephalosporins.

2. **Enzymatic creatinine assay:**
   - **Principle:** Creatinine -> creatine (creatininase) -> sarcosine + urea (creatinase) -> glycine + HCHO + H2O2 (sarcosine oxidase) -> colour (Trinder reaction, peroxidase + chromogen).
   - **Detection:** Absorbance at **546 nm** (Trinder) or **340 nm** (NADH-coupled).
   - **LOD:** ~0.05 mg/dL.
   - **Advantages:** More specific than Jaffe; less interference.
   - **Disadvantages:** More expensive reagents.
3. **HPLC (reference method):**
   - **Principle:** Reversed-phase or ion-exchange HPLC with UV detection at **234 nm** (creatinine absorption).
   - **LOD:** ~0.01 mg/dL.
   - **Advantages:** Highly specific; reference method (NIST SRM 967).
   - **Disadvantages:** Not routine; research/reference labs.
4. **IDMS (isotope dilution mass spectrometry):**
   - **Principle:** Gold-standard reference method for creatinine calibration.
   - **LOD:** <0.001 mg/dL.

## Optimal Urine Type for Measurement

- **24-hour urine** for creatinine clearance calculation and total excretion assessment (completeness check: expected 15-25 mg/kg/day).
- **Spot urine** for creatinine-normalised ratios (albumin/creatinine, protein/creatinine).
- Second morning void preferred for spot ratios (less variation).
- Stable at room temperature for 24-48 h; refrigerate for longer storage.

## Actual Gold Standard

**IDMS** is the ultimate reference method for creatinine standardisation. For clinical use, **enzymatic assays** (IDMS-traceable calibration) are preferred over Jaffe. Both are available on all major automated platforms. CV: 2-5% (Jaffe), 1-3% (enzymatic).

## Interferences in Measurement

| Interference | Effect | Mechanism |
|---|---|---|
| **Glucose (Jaffe)** | Positive bias | Reacts with alkaline picrate |
| **Acetoacetate (Jaffe)** | Positive bias | Ketone body interference |
| **Cephalosporins (Jaffe)** | Positive bias | Chromogen formation |
| **Bilirubin (Jaffe)** | Negative bias | Absorbs at 520 nm; subtractive |
| **Haemolysis** | Variable | Released proteins interfere |
| **Ascorbic acid (enzymatic)** | Negative bias | Inhibits Trinder reaction |
| **Dopamine, dobutamine** | Positive bias (enzymatic) | Interfere with H2O2 detection |

## Research Detection Methods

### *Spectroscopy Detection (UV-Vis / NIR)*

- **UV absorption:** Creatinine absorbs at **234 nm** (epsilon ~6,850 M-1cm-1). Direct UV quantification possible but non-specific in urine matrix.
  - LOD: ~0.5 mg/dL in buffer; urine requires HPLC separation.
- **Jaffe chromogen:** 520 nm (see above).

- **NIR:** Creatinine has weak NIR overtone bands (~2170 nm). Multivariate calibration with PLS gives LOD ~0.5 mg/dL in urine.

#### *Fluorescence Detection*

- Creatinine itself is weakly fluorescent (Ex ~240 nm / Em ~390 nm) – not useful.
- **Enzymatic fluorometric:** Sarcosine oxidase cascade produces H2O2 -> coupled with Amplex Red (Ex 571 / Em 585 nm). LOD: ~0.01 mg/dL.
- **3-Hydroxyflavone probes:** React with creatinine in alkaline conditions -> fluorescent product (Ex 370 / Em 520 nm). LOD: ~0.1 mg/dL.

#### *Raman Detection*

Creatinine has characteristic Raman peaks:

| Peak (cm-1) | Assignment |
|---|---|
| **~680** | Ring breathing |
| **~845-850** | C-N stretch |
| **~910** | Ring deformation |
| **~1400-1420** | CN stretch + CH3 deformation |
| **~1490** | C=N stretch |

- Conventional Raman: LOD ~5 mM (~56 mg/dL) – adequate for concentrated urine.
- SERS (Ag nanoparticles): LOD ~10-100 uM.
- Excitation: 532 or 785 nm.
- Sample prep: None (direct) or nanoparticle mixing.

#### *FTIR Detection*

| Band (cm-1) | Assignment | Notes |
|---|---|---|
| **~1492** | C=N stretch | Key diagnostic band (shifts with hydration) |
| **~1590** | NH bending | Overlaps with urea |
| **~1670** | C=O stretch | Ring carbonyl |
| **~2500-3200** | N-H stretches | Broad |

- ATR-FTIR can quantify creatinine in urine (the 1492 cm-1 band). LOD: ~5 mM in liquid urine.
- Hydration-dependent spectral changes documented (Rich & Sheringham, 2017).
- Sample prep: Direct ATR measurement or dried film.

#### *Voltammetry Detection*

- **Direct oxidation:** Creatinine oxidised at +0.8-1.0 V vs Ag/AgCl on modified electrodes (copper nanoparticles, MIP/graphene composites).
  - LOD: ~1-10 uM on nanostructured electrodes.
- **MIP-based electrochemical sensors:** Molecularly imprinted polymer for creatinine recognition.
  - LOD: ~0.5 uM.
  - Linear range: 1-1000 uM.
- **Sample prep:** Dilution in buffer; pH adjustment.

### Other Detection Technologies

1. **Capillary electrophoresis:** UV detection at 200-234 nm. LOD: ~0.5 uM. Fast separation.
2. **Paper-based colorimetric (Jaffe):** Picrate-impregnated paper strip + smartphone readout. LOD: ~0.5 mg/dL.
3. **Microfluidic enzymatic sensor:** Lab-on-chip with integrated enzymatic cascade. LOD: ~0.1 mg/dL.

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## Summary Table: Detection Methods Comparison

| Method | LOD | Key Parameter | Sample Prep | Strengths |
|---|---|---|---|---|
| **Jaffe reaction** | ~0.1 mg/dL | 520 nm | None | Universal, cheap |
| **Enzymatic assay** | ~0.05 mg/dL | 546 nm | None | More specific |
| **HPLC-UV** | ~0.01 mg/dL | 234 nm | Inject | Reference method |
| **IDMS** | <0.001 mg/dL | m/z | Extraction | Ultimate reference |
| **ATR-FTIR** | ~5 mM | 1492 cm-1 | None/dry | Reagent-free |
| **Raman** | ~5 mM | 680, 1490 cm-1 | None | Non-destructive |
| **SERS** | ~10-100 uM | 680 cm-1 | Nanoparticles | Enhanced sensitivity |
| **Voltammetry (MIP)** | ~0.5 uM | +0.8-1.0 V | Buffer dilution | Sensitive, selective |
| **Fluorometric (Amplex)** | ~0.01 mg/dL | Ex 571/Em 585 nm | Enzyme cascade | Very sensitive |

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## References


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